

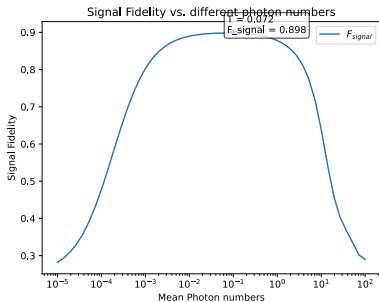
Update: Atom-Photon Gate in QuTiP + Short Lab Update

Learnt Zuka

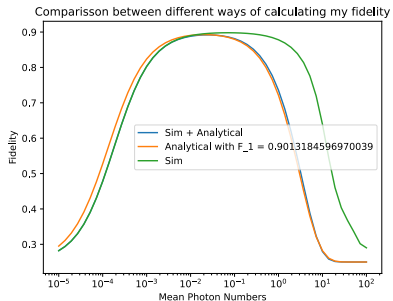
Master Thesis Progress Report

February 16, 2026

Results: Fidelity and Atomic Scattering vs Photon Number



Fidelity vs $\langle n \rangle$

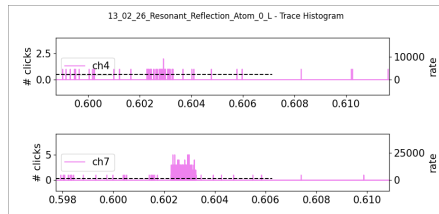


Comparison between different ways of
computing Fidelity

Reflection Resonant Light



Reflection with atom in $|1\rangle$ and
sending $|L\rangle$



Reflection with atom in $|0\rangle$ and
sending $|L\rangle$